

SOFIA LOYA

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EDUCATION

Carnegie Mellon University

Master of Mechanical Engineering

Pittsburgh, PA

May 2027

Texas A&M University

Bachelor of Science in Mechanical Engineering

GPA: 3.587/4.0

College Station, TX

May 2021

SKILLS

- Proficient in: SolidWorks, Creo, Visual Basic for Applications (VBA), JIRA, Agile, data analytics, Engineering fundamentals: solid mechanics, material science, heat transfer, mechanism design, manufacturing
- Skilled in: rapid prototyping (FDM technologies), PowerBI, Finite Element Analysis, and applied statistics
- Languages: Spanish

PROFESSIONAL EXPERIENCE

Dell Technologies | Dell Commercial Mobile Latitude

Mechanical Engineer II

Austin, TX

April 2023 – June 2025

- Managed 8 builds of materials (BOMs) with 200+ parts each and drove design changes to effectively reduce costs by 12M USD
- Increasing sustainable material content in part designs by 14% across three platforms
- Created weight reduction opportunities and decreased weight of single platform by 10%, exceeding marketing targets
- Improved internal specifications for all commercial mobile products to maintain competitiveness against market products
- Initiated and developed 'Recent Graduate CREO Learning' by detailing a CAD development path for new hires

Dell Technologies | Dell Commercial Mobile Rugged

Mechanical Engineer I

Austin, TX

June 2021 – April 2023

- Created engineering design documents to communicate all product features and design specifications to ODM
- Led cross-functional team for four weeks turning a \$240 million security business into a \$2 billion business
- Developed and filed intrusion patent to improve security of mobile products
- Built comparison interface for mobile platforms and competitors in Power BI
- Designed I/O port housing in CREO and analyzed mechanical structure in ANSYS to reduce testing time

Dell Technologies | Dell Desktop Precision

Mechanical Engineer Intern

Austin, TX

May 2020 – August 2020

- Developed housing for removable SSD cards in Creo for Precision tower quadrupling memory capacity
- Collaborated with engineering team to design module that increases capacity of Precision towers by 8 TB

GE Aviation | Test Enabling Hardware

Test Systems Engineering Intern

Cincinnati, OH

May 2019 – August 2019

- Gained sourcing experience and practiced communication with vendors
- Presented design reviews before lead engineers once a month to formulate final design model in NX
- Assisted Chief Engineers and Sub-Section Managers update engineering drawings database

Ethicon | Laparoscopic Surgical Products

R&D Intern

Cincinnati, OH

January 2019 – May 2019

- Created a test fixture for laparoscopic surgery simulation with my knowledge of mechanics and thermodynamics
- Drove design verification testing for 40 devices that passed 100% of requirements
- Utilized applied statistics to analyze test data and related it to physical phenomena for seven test methods